

MATH 4340/ COSC 4340 – Numerical Methods for Differential Equations

HW 1: Due by midnight on Saturday January 27, 2024.

Submit to WyoCourses a zipped folder containing a Jupyter notebook (both ipynb and HTML files).

Handwritten work should be included as a PDF or typed into the notebook as LaTeX.

1. [10 points] Determine Lipschitz constants for the following functions:

$$f(t, y) = 2y/t, \quad t \geq 1$$

$$f(t, y) = t - y^2, \quad |y| \leq 10.$$

2. [50 points] Let $t \in [0, 1]$ and $y(0) = 3$, and consider the initial value problem (IVP)

$$\frac{dy}{dt} = f(t, y) = -2y + 3e^t.$$

- (A). [5 point] Numerically integrate the IVP using the forward Euler (FE) method with $N = 4$ time steps: $t_i = t_0 + hi$, where $h = (T - t_0)/N$ and $i = 0, 1, \dots, N$. Make a plot that compares the true solution $y(t) = 2e^{-2t} + e^t$ to your predictions y_i . Plot $y(t)$ as a line and the y_i values as symbols or dots.
- (B). [5 points] Increasing N , the number of steps, is called “mesh refinement.” Repeat Problem 2A, but now use four choices of N , specifically $N \in [4, 16, 64, 256]$. Make a single plot that compares the predictions y_i to the true function $y(t)$ for all values of N .
- (C). [20 point] For each of these different choices of $N \in [4, 16, 64, 256]$, find the errors at the first and last time,

$$e_1 = |y(t_1) - y_1|, \quad E_N = |y(t_N) - y_N|,$$

and then make two plots that compare e_1 versus h and E_N versus h , respectively. Show them in a loglog scale using the Python function `loglog()` instead of `plot()`. Validate that the convergence rates are $e_1 \sim \mathcal{O}(h^2)$ and $E_N \sim \mathcal{O}(h)$, respectively. *Hint: Use `polyfit` to find the slopes as discussed in the first example Jupyter notebook.*

- (D). [20 points] Obtain a Lipschitz constant L and constant M so that the local and global errors for the FE method for this IVP satisfy the bounds,

$$e_1 \leq \frac{Mh^2}{2}, \quad E_N \leq \frac{Mh}{2L}(e^{L(T-t_0)} - 1),$$

and repeat Problem 2C but additionally plot these bounds. This should validate that the errors are in fact smaller than the bounds. *Hint: To compute M , you may use that $|y| \leq 3$.*

3. [40 points] You will now solve the IVP given in Problem 2 using the 2nd-order Taylor method. (Note that the 1st-order Taylor method is identical to Forward Euler.)

- (A). [10 point] Derive a 2nd-order Taylor method to solve the IVP using only knowledge of $f(t, y)$. *Hint: Recall that $\frac{d}{dt}f(t, y) = f_t + f f_y$, where f_t and f_y are partial derivatives with respect to t and y .*
- (B). [10 point] Numerically solve the IVP with $N = 10$ time steps using your 2nd-order Taylor method. Make a plot that compares your predictions y_i to both the true solution $y(t)$ as well as a numerical solution using the FE method (which you will also solve using $N = 10$).
- (C). [20 point] Design and implement an experiment to numerically validate that the local and global error for the 2nd-order Taylor method satisfy bounds of the form:

$$e_1 \leq ch^3, \quad E_N \leq \frac{ch^2}{L}(e^{L(T-t_0)} - 1)$$

for some constant $c > 0$. *Hint: See Problem 2C above.*